

Depth Super-Resolution Meets Uncalibrated Photometric Stereo

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ICCV 2017 Color and Photometry in Computer Vision Workshop

Outline

- 1 Introduction
- 2 Background
- 3 Methodology
- 4 Evaluation and Results
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Problem Statement

Example: RGB-D data from ASUS Xtion Pro Live



Input RGB image



Input depth

- + Good quality
- + High resolution

- Noisy & missing areas
- Low resolution

Goal



Input RGB image



Input depth



Refined depth

Objective:

Use high-resolution photometric clues in the RGB image to turn the low-resolution depth maps into a refined, high resolution one

Contribution

Propose a novel variational model to:

- disambiguate depth super-resolution through high-resolution photometric clues;
- disambiguate uncalibrated photometric stereo through low-resolution depth cues.

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Background

Depth Super-Resolution

$$\mathbf{z}_0^i = K\mathbf{z} + \varepsilon_z^i, \forall i \in \{1, \dots, n\}$$

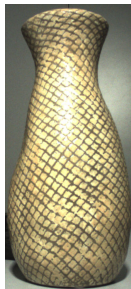
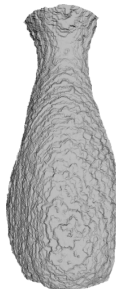
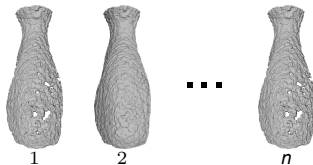
$\mathbf{z}_0^i$: input LR depth maps

$\mathbf{z}$: output HR depth map

K : down-sampling kernel

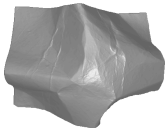
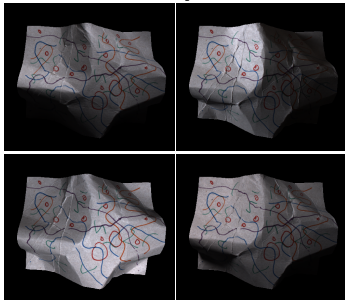
ε_z^i : noise $\sim \mathcal{N}(0, \sigma_z^2)$

$$\min_{\mathbf{z}} \mathcal{R}_z(\mathbf{z}) + \frac{1}{2n} \sum_{i=1}^n \|K\mathbf{z} - \mathbf{z}_0^i\|_{\ell^2}^2$$



Background

[Grosse et al., ICCV 2009]



Photometric Stereo

$$I^i = \rho \mathbf{l}^i \cdot \begin{bmatrix} \mathbf{n}(\mathbf{z}) \\ 1 \end{bmatrix} + \varepsilon_l^i, \forall i \in \{1, \dots, n\}$$

I^i : images under various lightings

$\mathbf{l}^i$: lighting vector $\mathbb{R}^4$

ρ : albedo / reflectance

$\mathbf{n}(\mathbf{z})$: surface normal

$$\min_{\mathbf{z}} \mathcal{R}_l(\mathbf{z}) + \frac{1}{2n} \sum_{i=1}^n \left\| \rho \mathbf{l}^i \cdot \begin{bmatrix} \mathbf{n}(\mathbf{z}) \\ 1 \end{bmatrix} - I^i \right\|_{\ell^2}^2$$

Background

Depth Super-Resolution

$$\mathbf{z}_0^i = K\mathbf{z} + \varepsilon_z^i, \forall i \in \{1, \dots, n\}$$

$\mathbf{z}_0^i$: input LR depths

$\mathbf{z}$: output HR depth

K : down-sampling kernel

ε_z^i : noise $\sim \mathcal{N}(0, \sigma_z^2)$

$$\min_{\mathbf{z}} \mathcal{R}_z(\mathbf{z}) + \frac{1}{2n} \sum_{i=1}^n \|K\mathbf{z} - \mathbf{z}_0^i\|_{\ell^2}^2$$

Photometric Stereo

$$\mathbf{I}^i = \rho \mathbf{l}^i \cdot \begin{bmatrix} \mathbf{n}(\mathbf{z}) \\ 1 \end{bmatrix} + \varepsilon_I^i, \forall i \in \{1, \dots, n\}$$

$\mathbf{I}^i$: images under various lightings

$\mathbf{l}^i$: lighting vector $\mathbb{R}^4$

ρ : albedo / reflectance

$\mathbf{n}(\mathbf{z})$: surface normal

$$\min_{\mathbf{z}} \mathcal{R}_I(\mathbf{z}) + \frac{1}{2n} \sum_{i=1}^n \|\rho \mathbf{l}^i \cdot \begin{bmatrix} \mathbf{n}(\mathbf{z}) \\ 1 \end{bmatrix} - \mathbf{I}^i\|_{\ell^2}^2$$

Background

Depth Super-Resolution

$$\mathbf{z}_0^i = K\mathbf{z} + \varepsilon_z^i, \forall i \in \{1, \dots, n\}$$

$\mathbf{z}_0^i$: input LR depths

$\mathbf{z}$: output HR depth

K : down-sampling kernel

ε_z^i : noise $\sim \mathcal{N}(0, \sigma_z^2)$

$$\min_{\mathbf{z}} \mathcal{R}_z(\mathbf{z}) + \frac{1}{2n} \sum_{i=1}^n \|K\mathbf{z} - \mathbf{z}_0^i\|_{\ell^2}^2$$

$$\text{Proposed Model: } \min_{\mathbf{z}} \frac{1}{2n} \sum_{i=1}^n \left\{ \|K\mathbf{z} - \mathbf{z}_0^i\|_{\ell^2}^2 + \lambda \left\| \rho \mathbf{l}^i \cdot \begin{bmatrix} \mathbf{n}(\mathbf{z}) \\ 1 \end{bmatrix} - \mathbf{l}^i \right\|_{\ell^2}^2 \right\}$$

Photometric Stereo

$$\mathbf{l}^i = \rho \mathbf{l}^i \cdot \begin{bmatrix} \mathbf{n}(\mathbf{z}) \\ 1 \end{bmatrix} + \varepsilon_l^i, \forall i \in \{1, \dots, n\}$$

$\mathbf{l}^i$: images under various lightings

$\mathbf{l}^i$: lighting vector $\mathbb{R}^4$

ρ : albedo / reflectance

$\mathbf{n}(\mathbf{z})$: surface normal

$$\min_{\mathbf{z}} \mathcal{R}_l(\mathbf{z}) + \frac{1}{2n} \sum_{i=1}^n \left\| \rho \mathbf{l}^i \cdot \begin{bmatrix} \mathbf{n}(\mathbf{z}) \\ 1 \end{bmatrix} - \mathbf{l}^i \right\|_{\ell^2}^2$$

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Methodology

With $(i, \star, \mathbf{p})$ the indices of images, channel and pixel,

$$I_{\star}^i(\mathbf{p}) = \rho_{\star}(\mathbf{p}) \mathbf{l}_{\star}^i \cdot \begin{bmatrix} \mathbf{n}(\mathbf{p}) \\ 1 \end{bmatrix} + \varepsilon_{\star}^i(\mathbf{p})$$

Methodology

With $(i, \star, \mathbf{p})$ the indices of images, channel and pixel,

$$\left. \begin{aligned} l_{\star}^i(\mathbf{p}) &= \rho_{\star}(\mathbf{p}) l_{\star}^i \cdot \begin{bmatrix} \mathbf{n}(\mathbf{p}) \\ 1 \end{bmatrix} + \varepsilon_{\star}^i(\mathbf{p}) \\ \mathbf{n}(\mathbf{p}) &= \frac{1}{d(\mathbf{z})(\mathbf{p})} \begin{bmatrix} f \nabla \mathbf{z}(\mathbf{p}) \\ -\mathbf{z}(\mathbf{p}) - \nabla \mathbf{z}(\mathbf{p}) \cdot (\mathbf{p} - \mathbf{p}^0) \end{bmatrix} \end{aligned} \right\}$$

f : focal length

$\mathbf{p}^0$: principal point

$d(\mathbf{z})$: normalizer

Methodology

With $(i, \star, \mathbf{p})$ the indices of images, channel and pixel,

$$\left. \begin{aligned} l_{\star}^i(\mathbf{p}) &= \rho_{\star}(\mathbf{p}) \mathbf{l}_{\star}^i \cdot \begin{bmatrix} \mathbf{n}(\mathbf{p}) \\ 1 \end{bmatrix} + \varepsilon_{\star}^i(\mathbf{p}) \\ \mathbf{n}(\mathbf{p}) &= \frac{1}{d(\mathbf{z})(\mathbf{p})} \begin{bmatrix} f \nabla \mathbf{z}(\mathbf{p}) \\ -\mathbf{z}(\mathbf{p}) - \nabla \mathbf{z}(\mathbf{p}) \cdot (\mathbf{p} - \mathbf{p}^0) \end{bmatrix} \end{aligned} \right\} \mathbf{A}^i(\mathbf{z}, \boldsymbol{\rho}, \mathbf{l}^i)^{\top} \begin{bmatrix} \nabla \mathbf{z} \\ \mathbf{z} \end{bmatrix} = \mathbf{b}^i(\boldsymbol{\rho}, \mathbf{l}^i) + \varepsilon^i$$

f : focal length

$\mathbf{p}^0$: principal point

$d(\mathbf{z})$: normalizer

Proposed Variational Model

Here we have:

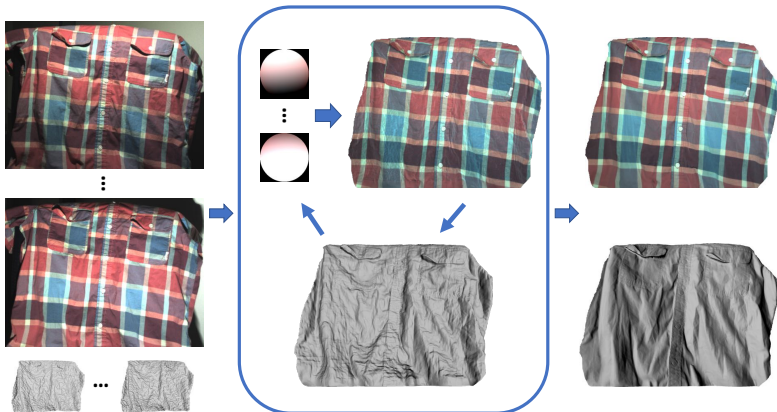
- depth super-resolution cue: $\mathbf{z}_0^i = K\mathbf{z} + \varepsilon_z^i, \forall i \in \{1, \dots, n\}$
- photometric stereo cue: $\mathbf{A}^i(\mathbf{z}, \boldsymbol{\rho}, \mathbf{l}^i)^\top \begin{bmatrix} \nabla \mathbf{z} \\ \mathbf{z} \end{bmatrix} = \mathbf{b}^i(\boldsymbol{\rho}, \mathbf{l}^i) + \varepsilon^i$

The final variational model is acquired from maximum likelihood:

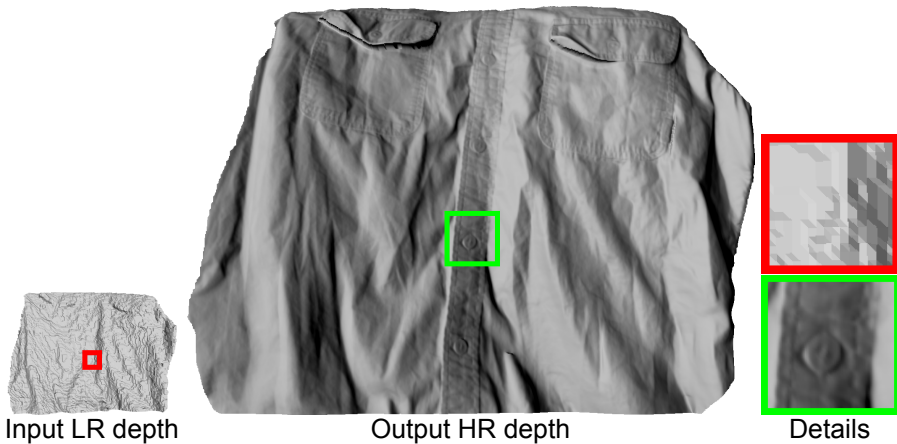
$$\min_{\mathbf{z}, \boldsymbol{\rho}, \{\mathbf{l}^i\}_i} \left\{ \sum_{i=1}^n \|\mathbf{K}\mathbf{z} - \mathbf{z}_0^i\|_{\ell^2}^2 + \lambda \sum_{i=1}^n \left\| \mathbf{A}^i(\mathbf{z}, \boldsymbol{\rho}, \mathbf{l}^i)^\top \begin{bmatrix} \nabla \mathbf{z} \\ \mathbf{z} \end{bmatrix} - \mathbf{b}^i(\boldsymbol{\rho}, \mathbf{l}^i) \right\|_{\ell^2}^2 \right\}$$

Alternating Optimization Workflow

$$\min_{\mathbf{z}, \boldsymbol{\rho}, \{\mathbf{l}^i\}_i} \left\{ \sum_{i=1}^n \|\mathbf{K}\mathbf{z} - \mathbf{z}_0^i\|_{\ell^2}^2 + \lambda \sum_{i=1}^n \left\| \mathbf{A}^i(\mathbf{z}, \boldsymbol{\rho}, \mathbf{l}^i)^\top \begin{bmatrix} \nabla \mathbf{z} \\ \mathbf{z} \end{bmatrix} - \mathbf{b}^i(\boldsymbol{\rho}, \mathbf{l}^i) \right\|_{\ell^2}^2 \right\}$$



Alternating Optimization Workflow



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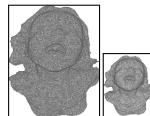
Synthetic Data



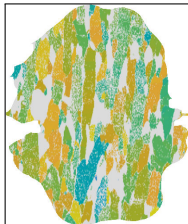
3D shape



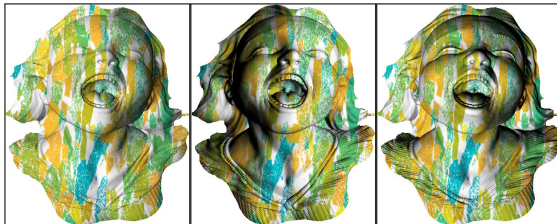
Ground truth HR depth



LR noisy depth



HR albedo map¹

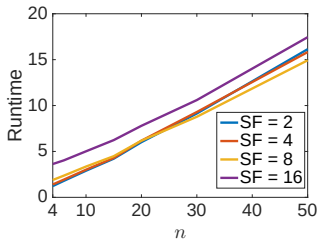
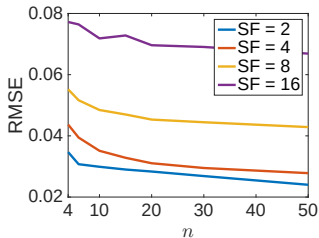
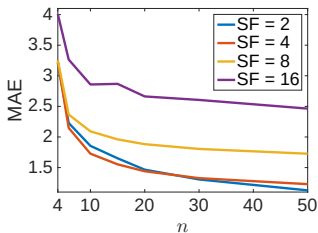


HR photometric stereo images

¹Source: <https://mtex-toolbox.github.io/files/doc/EBSDSpatialPlots.html>

Quantitative Evaluation

Number of images

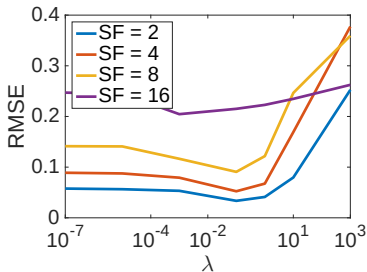
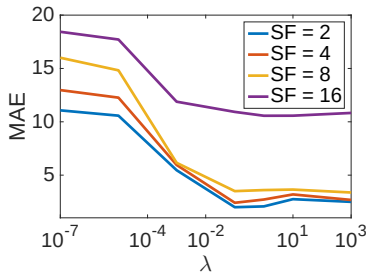


$n \in [10, 30]$ is a good compromise between accuracy and speed

Quantitative Evaluation

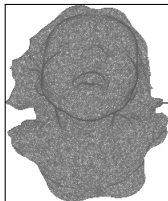
Parameter tuning

$$\min_{\mathbf{z}, \boldsymbol{\rho}, \{\mathbf{l}^i\}_i} \left\{ \sum_{i=1}^n \|\mathbf{K}\mathbf{z} - \mathbf{z}_0^i\|_{\ell^2}^2 + \lambda \sum_{i=1}^n \left\| \mathbf{A}^i(\mathbf{z}, \boldsymbol{\rho}, \mathbf{l}^i)^\top \begin{bmatrix} \nabla \mathbf{z} \\ \mathbf{z} \end{bmatrix} - \mathbf{b}^i(\boldsymbol{\rho}, \mathbf{l}^i) \right\|_{\ell^2}^2 \right\}$$



$\lambda \in [10^{-2}, 10^1]$ provide satisfactory results

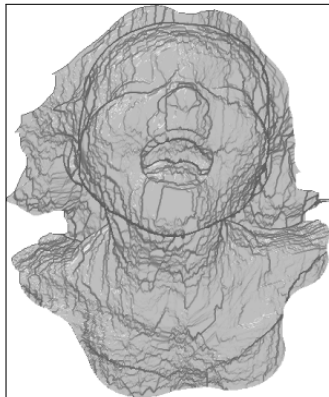
Quantitative Evaluation



RMSE = 0.0579

MAE = 65.7150

Input depth

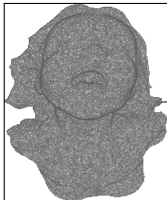


RMSE = 0.0728

MAE = 34.4129

Depth super-resolution with TV

Quantitative Evaluation



RMSE = 0.0579

MAE = 65.7150

Input depth

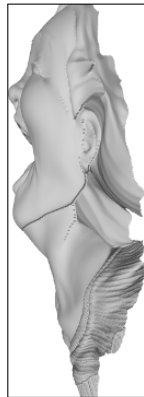


RMSE = 0.9199

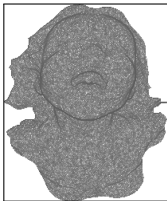
MAE = 41.8041

LDR Photometric Stereo

[Papadhimitri and Favaro, *IJCV* 2014]



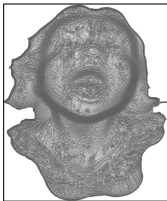
Quantitative Evaluation



RMSE = 0.0579

MAE = 65.7150

Input depth



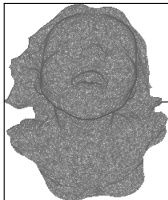
RMSE = 0.1655

MAE = 38.9316

RGBD-Fusion

[Or-Ei et al., CVPR 2015]

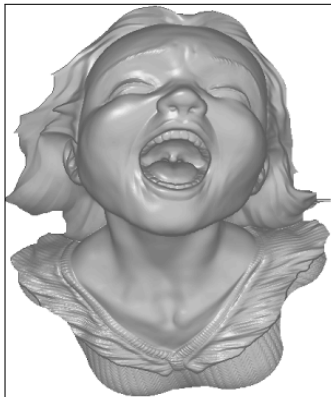
Quantitative Evaluation



RMSE = 0.0579

MAE = 65.7150

Input depth



RMSE = **0.0314**

MAE = **1.45280**

Ours

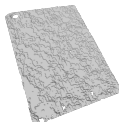
Quantitative Evaluation

	RMSE	MAE
Input	0.0579	65.7150
Depth SR	0.0728	34.4129
LDR PS	0.0919	41.8041
RGBD-Fusion	0.1655	38.9316
Ours	0.0314	1.45280

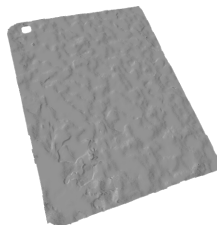
Qualitative Evaluation



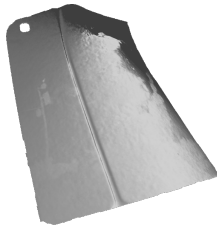
Input RGB image



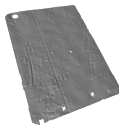
Input depth



Depth SR



LDR-PS



RGBD-Fusion



Ours

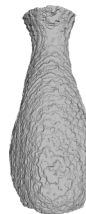
Qualitative Evaluation



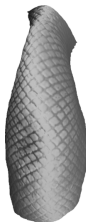
Input RGB image



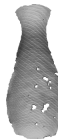
Input depth



Depth SR



LDR-PS



RGBD-Fusion



Ours

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Conclusions and Future work

- We proposed a novel variational framework for joint depth super-resolution and reflectance/light estimation
- Our method can be used out-of-the-box with common devices
- Theoretical analysis of this approach will be the next step

Data and codes are available on
<https://github.com/pengsongyou/SRmeetsPS>

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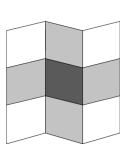
Computer Vision Group

Technical University of Munich

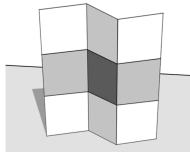


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Shape from shading ambiguity



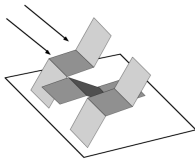
(a) An image



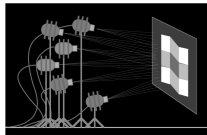
(b) A possible explanation



(c) painter's



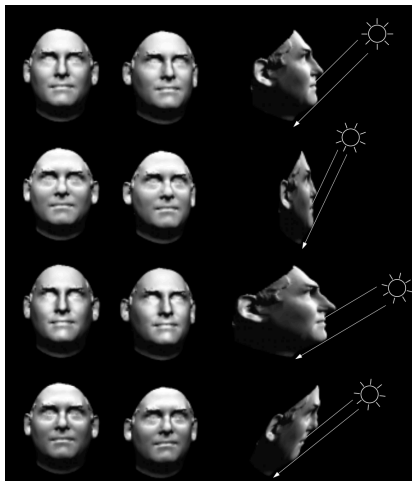
(d) sculptor's



(e) Lighting designer's

[Adelson and Pentland, 1996]

Generalized Bas-Relief (GBR)



[belhumeur et al., IJCV 99]